

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions
COURSE NAME: Cloud Computing and Big Data Analytics
COURSE CODE: CSA15

CO2: Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)

Dave's Taxonomy: Manipulation (Level 2) Question Count: 5 Total Marks: 25

Q1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how cloud auto-scaling and load balancing can handle this demand efficiently. How does this approach reduce downtime and improve user experience? What role do monitoring tools play in this setup? Relate your answer to a real-world e-commerce scenario. [5 Marks]

Answer:

Auto-Scaling and Load Balancing

Cloud auto-scaling automatically adds or removes compute instances (VMs or containers) based on real-time demand, so during a sudden spike in traffic the system provisions additional servers within minutes, and scales back down once demand subsides to save cost. Load balancers then distribute incoming requests evenly across all active instances, preventing any single server from becoming overloaded and ensuring consistent response times.

Reducing Downtime and Improving User Experience

By dynamically matching capacity to demand, auto-scaling prevents server crashes and slow response times that would otherwise occur under sudden load, directly reducing downtime. Load balancing adds fault tolerance: if one instance fails, traffic is automatically rerouted to healthy instances, so users experience uninterrupted service, faster page loads, and reliable checkout processes even during peak traffic.

Role of Monitoring Tools

Monitoring tools (e.g., AWS CloudWatch, Azure Monitor) continuously track metrics such as CPU utilization, request latency, and error rates, triggering auto-scaling policies when thresholds are crossed. They also provide real-time dashboards and alerts, allowing operations teams to detect anomalies and validate that scaling actions are actually resolving the load before problems affect users.

Real-World E-Commerce Scenario

During a Black Friday or Diwali sale, an e-commerce platform such as Amazon or Flipkart may see traffic increase tenfold within minutes. Auto-scaling groups behind an elastic load balancer spin up additional web and application servers automatically, while monitoring tools track checkout success rates; this ensures the site remains available and responsive throughout the sale instead of crashing under the surge.

Q2. An organization wants to store large volumes of backup data securely in the cloud. Discuss how cloud storage solutions ensure durability and data redundancy. How can encryption and access control protect sensitive backup data? Explain the cost advantages compared to traditional storage systems. Provide an example of a disaster recovery use case. [5 Marks]

Answer:

Durability and Redundancy

Cloud storage services (e.g., Amazon S3, Azure Blob Storage) achieve high durability, often 99.999999999% (11 nines), by automatically replicating data across multiple physical disks, and frequently across multiple datacenters or geographic regions. This redundancy means that even if a disk, server, or entire datacenter fails, the data remains intact and recoverable from another replica.

Encryption and Access Control

Sensitive backup data is protected using encryption at rest (data stored on disk is encrypted, e.g., AES-256) and encryption in transit (TLS during upload/download), so that even if storage media is stolen, the data is unreadable without keys. Access control mechanisms such as Identity and Access Management (IAM) policies, role-based access control (RBAC), and bucket/object-level permissions ensure only authorized users or applications can read, write, or delete backup data.

Cost Advantages

Traditional storage systems require large upfront capital expenditure on physical storage arrays, along with ongoing costs for power, cooling, and maintenance, often provisioned for peak capacity that goes underused. Cloud storage uses a pay-as-you-go model where the organization pays only for the storage actually consumed, with tiered pricing (e.g., cold storage/archive tiers for infrequently accessed backups) further reducing cost without requiring hardware ownership.

Disaster Recovery Use Case

An organization can configure cross-region replication so backup data stored in one cloud region is automatically copied to a geographically distant region. If a natural disaster or regional outage destroys the primary datacenter, the organization can restore operations from the replicated backup in the secondary region, minimizing data loss and downtime.

Q3. A mobile app company plans to deploy its application globally. Explain how content delivery networks (CDNs) improve performance and reduce latency. How do edge servers enhance user experience across regions? Discuss how cloud providers support global availability. Relate this to a real-time streaming or gaming application. [5 Marks]

Answer:

CDNs and Latency Reduction

A Content Delivery Network caches static and frequently accessed content (images, videos, scripts) on servers distributed across multiple geographic locations, so users retrieve content from a nearby server rather than a single origin server that may be far away. This significantly reduces round-trip time and page/content load latency for users anywhere in the world.

Role of Edge Servers

Edge servers are CDN nodes positioned physically close to end users in various regions; they serve cached content locally and can even perform lightweight compute (edge computing) such as request routing or content personalization. Because processing happens near the user rather than at a distant central datacenter, edge servers reduce delay, improve responsiveness, and provide a smoother experience even for users in regions far from the company's primary cloud region.

Cloud Provider Support for Global Availability

Major cloud providers (AWS, Azure, Google Cloud) operate data centers and edge points-of-presence across many continents, offering global load balancing that automatically routes each user's request to

the nearest healthy region. They also provide multi-region deployment tools, geo-DNS routing, and global CDN integrations (e.g., Amazon CloudFront, Azure CDN) that let a company deploy once and serve users worldwide with consistent performance.

Real-Time Streaming/Gaming Example

A live video streaming platform or online multiplayer game relies on ultra-low latency; using a CDN with edge servers ensures a viewer in Chennai and one in London both receive the video stream or game data from a nearby edge node instead of a single origin server, preventing buffering delays or lag and providing a smooth, synchronized experience for all players/viewers globally.

Q4. A financial institution requires strict compliance and data security in cloud usage. Explain how cloud providers support regulatory compliance and auditing. What role do logging, monitoring, and encryption play in security? How can the company ensure data sovereignty requirements are met? Give an example involving sensitive financial transactions. [5 Marks]

Answer:

Regulatory Compliance and Auditing

Cloud providers offer built-in compliance certifications (e.g., PCI-DSS, ISO 27001, SOC 2, GDPR-aligned controls) and provide compliance management tools (e.g., AWS Artifact, Azure Compliance Manager) that give financial institutions audit reports and evidence needed to satisfy regulators. They also support configurable audit trails that record every access and configuration change, which auditors can review to verify adherence to financial regulations.

Logging, Monitoring, and Encryption

Detailed logging (e.g., AWS CloudTrail, Azure Activity Log) records every API call and user action, creating a tamper-evident trail essential for forensic investigation and regulatory audits. Continuous monitoring detects anomalous behavior such as unauthorized access attempts in real time, triggering alerts. Encryption, both at rest and in transit, ensures that even if data is intercepted or storage is breached, sensitive financial information such as account numbers or transaction details remains unreadable to attackers.

Ensuring Data Sovereignty

The institution can select specific cloud regions to ensure customer financial data is stored and processed only within the country or jurisdiction mandated by local financial regulations (e.g., RBI guidelines in India requiring certain data to remain within national borders). Cloud providers allow region-locking of storage and compute resources, along with contractual data residency guarantees, to help meet these data sovereignty requirements.

Example: Sensitive Financial Transactions

A bank processing online fund transfers can encrypt transaction data end-to-end, log every transaction request with timestamps and user IDs for audit purposes, and restrict processing to an in-country cloud region to comply with data residency laws, while monitoring tools flag unusual transaction patterns that may indicate fraud, satisfying both security and regulatory obligations.

Q5. A development team wants to adopt containerization for application deployment. Explain how containers differ from virtual machines in terms of performance and resource usage. How does orchestration (like Kubernetes) simplify deployment and scaling? Discuss portability benefits across different cloud environments. Provide

an example of a microservices-based application. [5 Marks]

Answer:

Containers vs Virtual Machines

Virtual machines each include a full guest operating system running on top of a hypervisor, making them relatively heavyweight, slower to boot (minutes), and consuming more CPU/memory overhead. Containers, by contrast, share the host operating system's kernel and package only the application and its dependencies, making them lightweight, able to start in seconds, and far more efficient in resource usage, allowing many more containers than VMs to run on the same hardware.

Role of Orchestration (Kubernetes)

As the number of containers grows, manually managing their deployment, networking, and scaling becomes impractical. Kubernetes automates container deployment by scheduling containers onto available nodes, automatically restarting failed containers (self-healing), scaling the number of container replicas up or down based on load, and managing service discovery and load balancing between containers, greatly simplifying operations for large-scale applications.

Portability Benefits

Because a container packages the application with all its dependencies into a single portable image, it runs consistently across any environment that supports the container runtime, whether that is a developer's laptop, an on-premises server, or any major cloud provider (AWS, Azure, GCP). This eliminates the 'it works on my machine' problem and allows the team to move workloads between cloud providers or hybrid environments without rewriting the application.

Microservices Example

An e-commerce platform can be built as a microservices application where separate containers handle distinct functions such as user authentication, product catalog, shopping cart, and payment processing. Kubernetes orchestrates these containers, independently scaling the payment service during checkout spikes while keeping the catalog service running at normal capacity, and allows each microservice to be updated or redeployed independently without affecting the rest of the application.

Total Marks Awarded: 25 / 25